



Cisco Dwara

can combine multiple classes in remote areas and teach them complex concepts as if he or she were right in the classrooms. The students can also ask real-time questions as if the teacher was in their classroom. The cloud-driven solution also enables access to hosted resources. Since the device acts as a router, Wi-Fi access point, computer and projection device, it also allows multiple students using devices like the Akash tablet to share a single Internet connection and access other well-known education portals.

The solution is a high-quality package, with international compliance certified under FCC, CE and UL Labs. It is rugged and can work even in rural schools and can handle erratic power conditions, according to the company.

Aravind Sitaraman, president, Inclusive Growth, Cisco, says, "We are very proud to have developed this product from our site in India, realising our vision to bring cost of technology in education down to a very affordable \$1 per child per month."

Using 'voice' as a universal medium of info and service delivery

Social tags: Speech-based access to services, information

Tech tags: Mobile phones, voice biometrics, vernacular speech recognition

Despite the deep penetration of mobile networks, a large illiterate and semi-literate population of India is still unable to access information and services on the phone, because they cannot read or respond to SMS text.

A spark from Prof. Jhunjhun-

wala of IIT Madras sent mobile VAS entrepreneurs Ravi Saraogi and Umesh Sachdev along the social route; and after extensive field research they realised that the only way information and services could be delivered across

India's socioeconomic spectrum is using 'voice'—the only universal medium of communication and the most intuitive way of interacting with a phone. However, traditional interactive voice response (IVR) systems can be long, unintuitive and inhuman, putting off many people. What the entrepreneurs had in mind was quite unlike such methods; they were focused on 'speech recognition,' which enables natural, dynamic conversations between the application and the user.

"When looking at the various technologies that work on basic feature phones, we resolved that speech technology was the best option. In a country like India, nearly 40 per cent of the country is illiterate, and cannot interact with text-based communication. Moreover, up to 90 per cent of the country is not English literate. Most phones in India are incompatible with local language typography. Therefore up to 90 per cent of the country would not benefit from SMS-based marketing and outreach. Speech recognition, on the other hand, can communicate with people in their local languages over voice—the medium in which people are most comfortable interacting with their mobile phones," says Sachdev.

So Saraogi and Sachdev came up with an enterprise-class mobility platform that uses vernacular speech recognition and voice biometrics to enable automatic information delivery and capture to people on their mobile phones.

Uniphore builds mobility solutions using two primary technologies: multilingual speech recognition and voice

biometrics. Speech recognition enables businesses to deploy mobile solutions that understand and carry out the user's naturally spoken commands. Uniphore's technology interprets and responds to the particular characteristics of 14 Indian languages and over a hundred dialects. The speech-based applications use background noise cancellation, semantic interpretation and grammar specification to offer the most state-of-the-art products and services. The result is extremely high voice recognition accuracy that enables natural, human-like conversations and satisfying mobile interactions.

Uniphore's voice biometrics system captures the unique characteristics of an individual's voice in order to provide businesses with a secure and efficient speaker verification system. To enrol in the system, the user creates a 'voiceprint' by speaking a certain set of words. The voiceprint is as unique to the users as their fingerprint, consisting of 32 different elements of the user's speech pattern—both behavioural and physiological. It is used as the digital representation of their voice and as a standard of comparison when they call in the future.

"Using these technologies, our solutions enable businesses to send outbound calls to their constituents/customers in their local languages. The user can authenticate their identity through voice biometrics, then speak interactively with the speech recognition system to get the information they require. All the responses of the users are converted to text on the backend, and entered into the enterprise IT systems for analysis," says Sachdev.

Today, Uniphore has the widest deployment of speech recognition applications in the country, with nearly two million end users. It is being used by NGOs, governments and enterprises to reach out to people in every corner of India.

The Agriculture department of the government of Tamil Nadu, which is in a sanctioned partnership with RTBI, has taken Uniphore's solution and customised it to deliver specific